Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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1. Introduction

This report details an end-to-end analytics project that was completed for a publishing analytics team that helps online merchants and bookshops. The goal was to create a predictive model that uses catalogue data to estimate book prices and identify the elements that influence customer ratings and book prices within a selected group of popular genres.

Books to Scrape (books.toscrape.com), a sample e-commerce website designed especially for web scraping practice, provided the data. Eight different genres were chosen for extraction in order to guarantee a thorough and specialised study. The research adheres to the typical data science pipeline, which includes data collection, refinement, visual exploration, predictive modelling, and the conversion of results into useful business suggestions.

GitHub link: https://github.com/Tsakane05/Data-Science-Assignment.git

2. Methodology

2.1 Data collection

Python's requests and BeautifulSoup libraries were used to programmatically scrape Books to Scrape in order to collect data. The scraper was made to go to particular genre categories (Fiction, Mystery, Romance, Science Fiction, History, Young Adult, Poetry, and Travel) instead of scraping the entire catalogue. It collected all accessible titles by automatically handling pagination within each category.

A total of 215 items were gathered. The following characteristics were taken out of each book:

- Title, Price (GBP), and Star rating (1–5)
- Stock status and available quantity
- Genre category
- Product description
- Universal Product Code (UPC), pre-tax price, post-tax price, tax amount, and number of reviews

The raw data was stored as books_raw_v3.csv. To avoid server overload and disruptions, each request had a slight delay and strong error handling.

2.2 Data refinement

A Pandas DataFrame was filled with the scraped dataset. Blank strings were used to substitute missing descriptions, and pd.to_numeric was used to force columns containing numerical data (price, rating, quantity, tax, and reviews); rows with invalid prices or ratings were eliminated.

To improve the modelling dataset, four more engineering features were created:

- title_length — character count of the title

- description_length — character count of the description
- description_word_count — number of words in the description
- has_description — binary indicator of whether a description exists

The refined table was saved as books_clean_v3.csv.

2.3 Visual exploration

Using a unique pink and purple colour scheme, six charts were created with Matplotlib and Seaborn to examine the pricing behaviour of the catalogue:

1. Price distribution histogram
2. Rating frequency bar chart
3. Price boxplot across star ratings
4. The top genres by average price in a horizontal bar chart
5. Numerical feature correlation heatmap
6. Price boxplot by stock status

During the modelling phase, a seventh chart (actual vs. expected pricing) was created.

2.4 Predicative modelling

In Scikit-learn, a linear regression model was put together. Price was the goal variable, and the predictors were divided into two categories:

- Numerical: amount, rating, length of title, length of description, and number of reviews
- Categorical: stock status, genre

The numerical predictors were standardised via a preprocessing pipeline, whereas the categorical predictors were one-hot encoded. Using a fixed seed, the data was divided 80/20 into training and testing subsets. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R2 coefficient were used to gauge the quality of the model.

3. Results

3.1 Catalogue observations

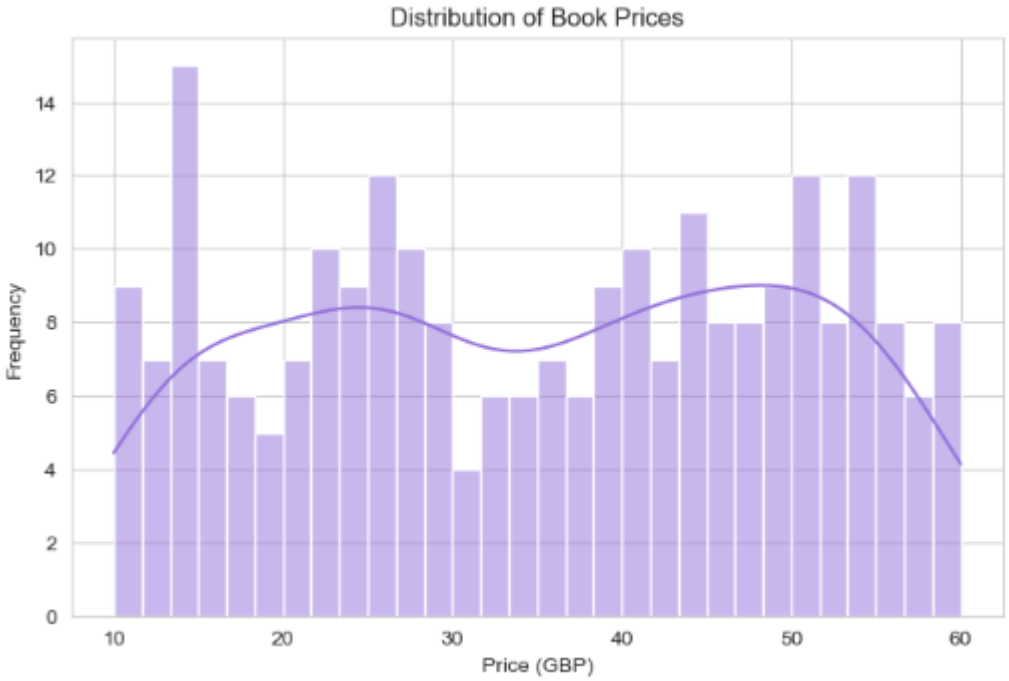


Figure 1: Shows how book prices are distributed throughout the catalogue. The bulk of titles are priced between £10 and £60, according to the histogram, which has a right-skewed price distribution.

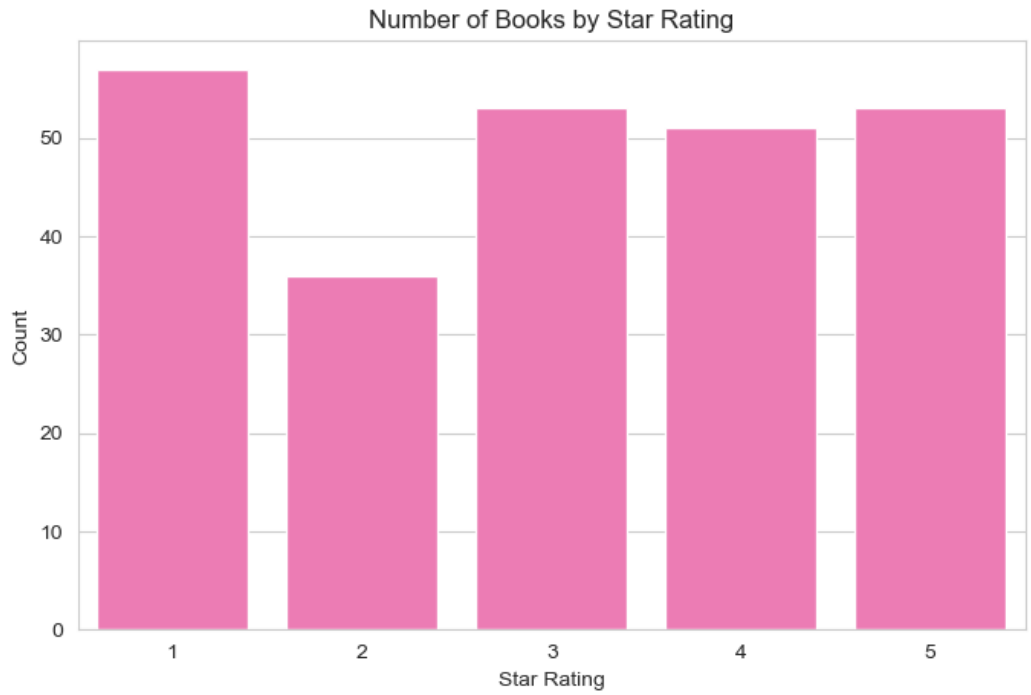


Figure 2: Book Count by Star Rating. There isn't a particular rating that dominates the chosen genres; instead, ratings are dispersed rather evenly over the 1–5 range.

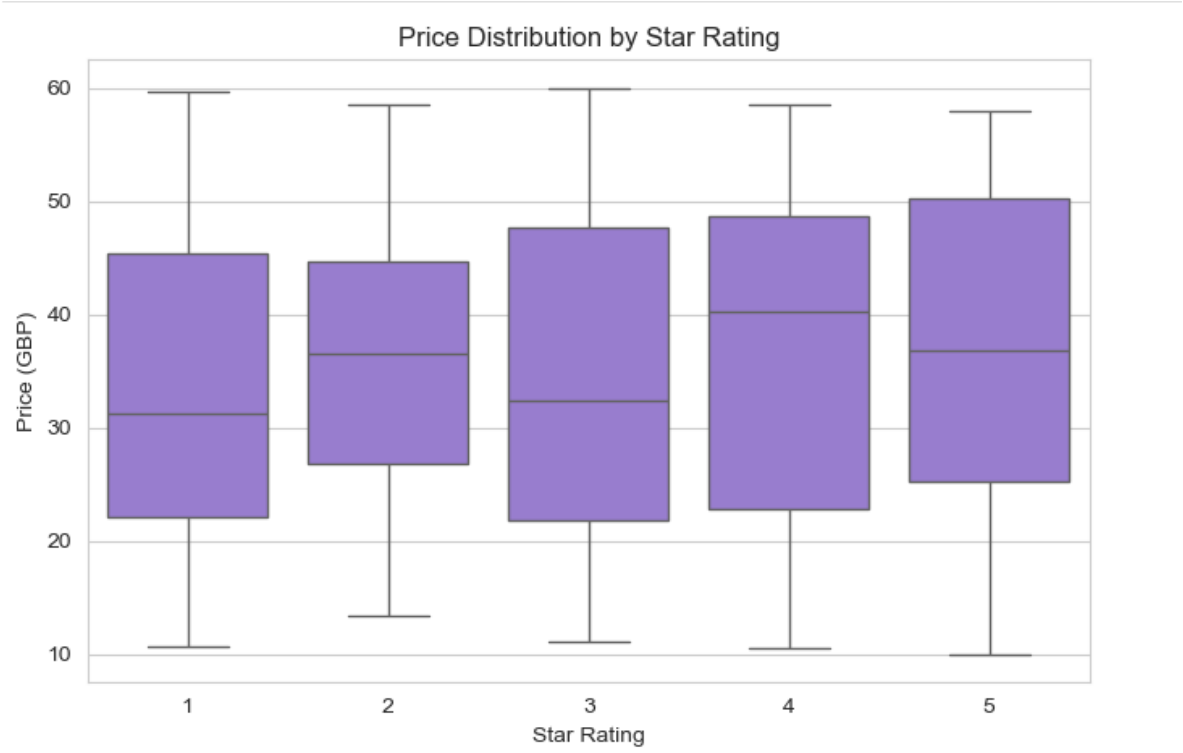


Figure 3: Price Distribution by Star Rating. Rating is a poor predictor of price, as the boxplot shows that median costs are comparable across rating categories.

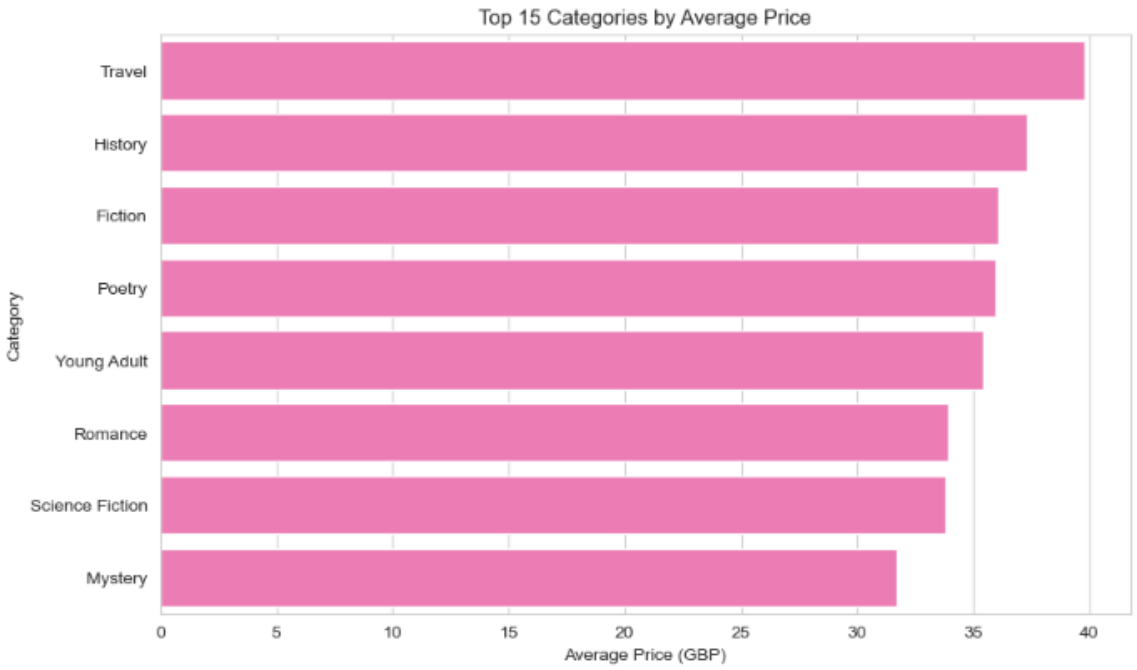


Figure 4: Top Genres by Average. There are distinct pricing trends based on genre: some genres are always at the top of the price range, while others are grouped below the baseline.

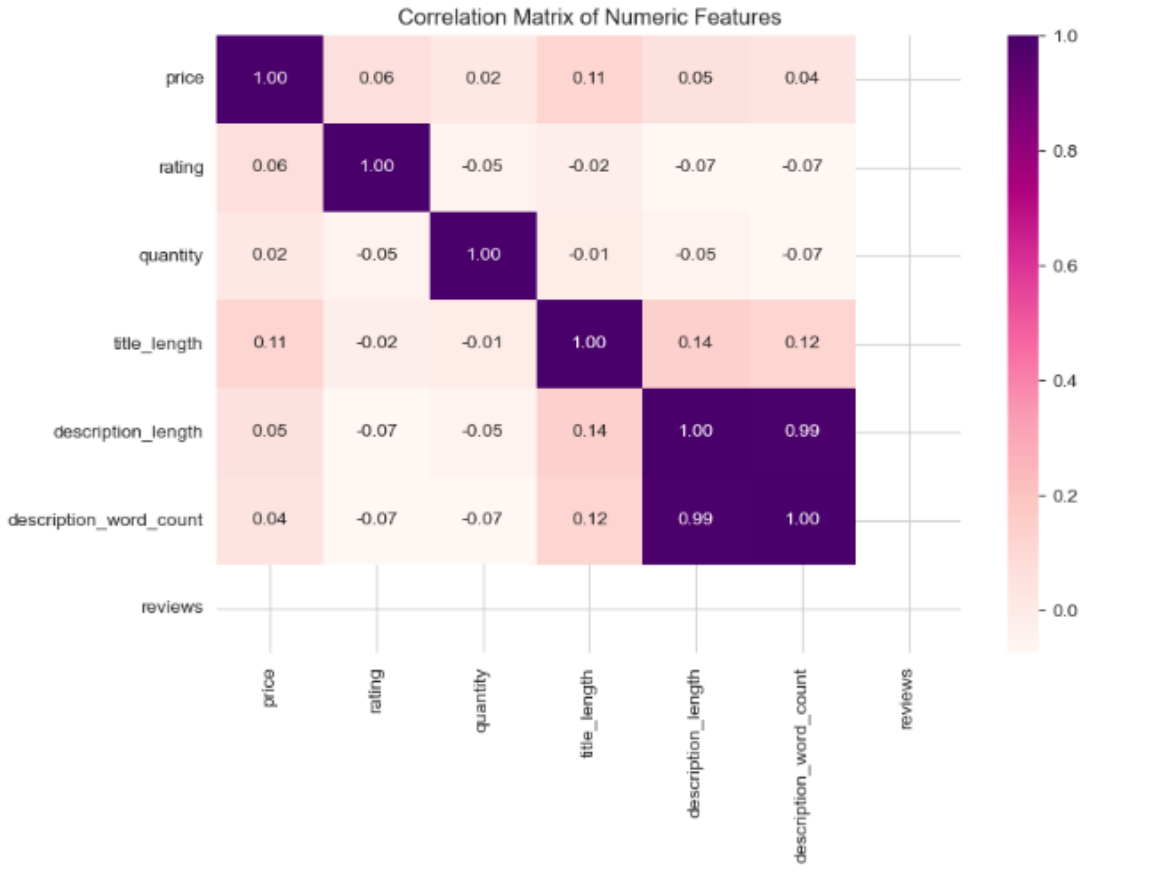


Figure 5: Numerical Features Correlation Matrix. The model's poor prediction accuracy is hinted at by the fact that the majority of numerical features have almost no link with price.



Figure 6: Price Distribution by Stock Availability (Figure 6). The price distributions of books that are in stock and those that are out of stock are almost comparable, indicating that stock status is not a significant pricing influence.

3.2 Model outcomes

On the test set, the Linear Regression model yielded the following outcomes:

Metric	Value	Interpretation
MAE	£15.49	The average difference between predictions and actual prices is £15.49.
RMSE	£17.33	As might be predicted with a poor model, there are larger errors.
R^2	-0.085	When compared to just forecasting the average price, the model performs somewhat worse.



Figure 7: Actual and predicted book prices are shown. The model's poor fit is visibly confirmed by the large dispersion of pink dots away from the perfect-prediction line.

3.3 Coefficient highlight

Categorical characteristics that reflect certain genres had the biggest coefficients in absolute value, indicating that a book's genre significantly influences the model's price forecast in relation to the baseline. Nearly nothing was provided by numerical predictors like rating and description length.

4. Discussion

The findings unequivocally demonstrate that the product-page attributes in this carefully selected genre sample cannot be used to accurately determine price. The negative R2 of -0.085 indicates the constraints of the underlying data and is not a sign of coding failure.

This result can be explained by several factors:

1. High-value elements missing: Demand, format (hardcover vs. paperback), publisher, and author renown all affect real retail prices; none of these factors are mentioned on the listing page.
2. Sandbox dataset: Books to Scrape is intended for scraping practice; instead of using an actual economic process, the pricing may be set arbitrarily.
3. Unsuitable linear assumption: The characteristics may have a non-linear relationship with pricing, necessitating the use of tree-based models such as Random Forest or Gradient Boosting.
4. Weak signal: There is hardly little association between price and even the best numerical features.

The pipeline itself was correctly implemented and repeatable, despite the model's poor predictive performance. A cleaner, more targeted dataset was obtained by extracting particular genres, demonstrating that product-page attributes lack the signal required for precise price prediction even after adjusting for genre.

5. Conclusion

For a carefully chosen collection of genres on the Books to Scrape database, a full data science pipeline was provided. Product-page features alone do not contain enough signal to explain price volatility, as demonstrated by the Linear Regression model's inadequate prediction power despite its clean training and evaluation.

6. Recommendations

1. Use price benchmarking based on genre. Retailers should use genre as the main anchor for pricing decisions because category dominates the model's signal.
2. Don't base prices on ratings. There is hardly any relationship between pricing and star ratings.
3. Extend the range of features. Future versions should include the author, publisher, format, number of pages, and year of publication.
4. Try other models. When paired with K-fold cross-validation, Random Forest, Gradient Boosting, or XGBoost may be more effective in capturing non-linear correlations.
5. Consider the model to be exploratory. For production pricing considerations, it is not yet appropriate.

7. Limitations

- The dataset does not accurately reflect actual market behaviour because it is a sandbox.
- Non-linear interactions cannot be captured by linear regression.
- Overfitting on small samples may result from one-hot encoding of genres, which increases feature dimensionality.

8. Biliograghy

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